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Real-rock microfluidic platform for quantifying chemical dissolution and mechanical erosion in a multiphase environment

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Fluid–rock interactions involving chemical dissolution, mechanical erosion, and multiphase flow are central to a wide range of geological and engineering processes, yet they remain poorly understood due to the lack of integrated *in situ* observation tools. Existing methods often compromise between spatial resolution and temporal dynamics. Here, we develop a real-rock microfluidic platform that enables simultaneous visualization and quantification of erosion dynamics in multiphase reactive systems. The platform integrates fluorescence microscopy, micro-particle image velocimetry, and ion chromatography to monitor the coupled evolution of solid–liquid–gas interfaces and flow velocity fields at micrometer-scale resolution. Microfluidic chips fabricated directly from limestone preserve natural mineral heterogeneity, and the platform enables direct observation of rock surface evolution and multiphase flow behavior. This facilitates decoupled analysis of chemical dissolution and mechanical erosion—two processes often difficult to isolate in traditional systems. Using this system, we investigate erosion during acid–rock interactions and identify a transition between two regimes—transport-limited and reaction-limited—controlled by CO_2 bubble mobility. In the transport-limited regime, immobile bubbles confine flow to thin films, enhancing dissolution and particle detachment. In the reaction-limited regime, surface-adhered bubbles shield reactive areas and reduce shear stress, suppressing erosion. We derive scaling laws that distinguish chemical and mechanical erosion rates and validate a theoretical model for the critical Péclet number marking the regime transition. This study advances understanding of erosion under multiphase flow and introduces a versatile experimental framework for probing pore-scale reactive transport. The platform can be extended to other rock types and fluids, offering a powerful tool for studying geochemical, physical, and biological processes in complex subsurface environments.

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1. Introduction

Across various scales of geological processes, fluid–fluid and fluid–rock interactions are commonplace. Phenomena such as diagenesis, mineralization, ore formation, and rock weathering are all outcomes of chemical and/or mechanical interactions between disparate fluid and solid phases.[1–4] Among these interactions, erosion plays a pivotal role in a range of natural and engineering processes, including rock denudation,[5,6] glacial abrasion,[7,8] the global carbon cycle,[4,9] CO_2 geological sequestration,[10,11] metal corrosion,[12,13] and the mobilization of toxic elements.[14,15] For example, carbonate rocks preferentially chemically react with CO_2-acidified water. The remaining unreacted mineral particles are then mechanically eroded. The combination of chemical dissolution and mechanical erosion accelerates the total erosion process and further impacts landscape evolution,[16] the global carbon cycle,[4,17] and the long-term safety of geological carbon sequestration.[18] This erosion is a pore-scale process recognized to be affected by hydrodynamics and reactive transport.[19] Fractures, ubiquitous in the Earth's subsurface, act as preferential flow pathways because of their high permeability, and thus typically control fluid migration and solute transport in fractured rocks.[19,20] In the Earth's critical zone, fractures control the availability of water for rock erosion and hence the development of the regolith layer.[20]

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The erosion of fractured media determines the evolution of fracture geometry and the transport properties.

Subsurface processes usually involve the flow of multiple fluids such as CO_2, water, and air.[21,22] For example, during the injection of CO_2 into deep saline aquifers, carbonic acid forms at the interface between the supercritical CO_2 ($scCO_2$) phase and resident brine, leading to dissolution of the surrounding rock matrix. The development of dissolution channels is further modulated by the presence of multiphase fluids, which reshape flow pathways and alter dissolution patterns.[23,24] Similarly, acid stimulation of carbonate formations through HCl injection can also create a multiphase reactive system, depending on reservoir pressure and acid concentration.[25,26] When the CO_2 production exceeds the dissolution capacity of the aqueous phase, a separate CO_2 phase emerges.[27,28] Accumulated gas has been shown to act as a retarding phase, thereby increasing the penetration depth of acid into the formation. However, most previous investigations of erosion dynamics were restricted to single-phase flow, neglecting the critical role of multiphase interactions in controlling transport and reaction processes.

Erosion in multiphase environments is much more complex due to the simultaneous presence of fluid–solid and fluid–fluid interactions, which give rise to dynamic interfacial phenomena.[29,30] Gas bubbles generated during reactive flow can significantly alter the flow paths, reduce the exposure of reactive surfaces, or modulate local shear stress. Furthermore, the reconfiguration of flow channels and increased near-wall velocities may promote localized dissolution and enhance the transport of detached particles.[31,32] Therefore, the erosion process is highly sensitive to the evolving distribution and evolution of gas–liquid interfaces, which manifest as transient microscale instabilities. The erosion of rock samples under multiphase flow conditions thus involves not only long-term dissolution of soluble minerals, but also instantaneous detachment of inert grains and rapid interface restructuring. Capturing these interrelated processes demands high-resolution experimental tools capable of real-time observation of erosion, bubble dynamics, and flow behavior. To date, such comprehensive and quantitative investigations remain limited.

The understanding of erosion dynamics has largely been informed by flow-through experiments using pure minerals such as gypsum,[33,34] halite,[35,36] potassium dihydrogen phosphate (KDP),[37] and calcite.[28,38] These studies typically focus on chemical dissolution. Studies focused on particle transport show that the mechanical erosion of non-cohesive grains depends primarily on shear stress.[31,39,40] For natural rocks, which contain various mineral particles and cementing minerals, erosion is further influenced by heterogeneous mineral composition. Core-flooding experiments with CT imaging have demonstrated that preferential dissolution of calcite creates alteration zones, leaving less reactive minerals more vulnerable to subsequent mechanical removal.[19,41] In rocks with high mineralogical heterogeneity, this process becomes increasingly significant.[42,43] However, mechanical erosion remains challenging to isolate and quantify, especially in experiments using pure minerals where dissolution dominates and detachment may not occur at all.

Due to the intricate coupling of physicochemical processes and their rapid temporal evolution, deciphering erosion dynamics in multiphase settings requires advanced experimental approaches capable of resolving spatial, temporal, and compositional changes in the pore structure. Traditional core-flooding methods rely on before-and-after imaging, offering limited insight into real-time interface evolution. For instance, X-ray CT offers spatial resolution down to 1.3 μm with temporal sampling intervals on the order of minutes.[14] AFM improves spatial resolution to the nanometer scale but is constrained to an imaging speeds of about 16 seconds per frame with a narrow field of view of 20 μm × 20 μm.[16] The latest imaging improvements that combine micro-CT setup with synchrotron beamlines allow for probing the unsteady flow field throughout the pores at a time resolution of 0.25 seconds.[44] Despite these improvements, such methods still cannot adequately capture rapid interface changes, particularly the behavior of undissolved mineral particles during erosion in high-velocity fractures where relevant phenomena unfold on millisecond or shorter timescales.

To overcome these limitations, microfluidic systems fabricated from natural rock samples provide a promising approach to investigate the erosion dynamics.[21,45–48] These platforms preserve native mineral textures and provide a direct observation of the physicochemical interactions between water and rock.[43] When combined with advanced metrological techniques, such systems enable high optical accessibility to multiphase reactive transport and allow for quantitative characterization of transport mechanisms,[49,50] chemical reactions,[51–53] interfacial processes,[54,55] and mixing phenomena[50,56] within the pore space. These advances have made it possible to directly visualize multiphase reactive transport in heterogeneous media, including dynamic evolution of gas–liquid interfaces and particle migration at the pore scale.[39,57] Nevertheless, the quantitative separation and measurement of mechanical erosion in these systems remains largely unexplored.

To address this gap, we present a novel experimental framework that integrates real-rock microfluidics with multiple optical imaging modalities and ion chromatography to achieve real-time quantification of erosion in multiphase environments. Our system achieves a spatial resolution of 1.3 μm and a temporal resolution of 1.0 ms, allowing direct visualization of fluid–fluid–solid interfaces and particle dynamics. This combination of technologies enables the characterization of physical parameters that are otherwise difficult to measure, including flow fields, interfacial stresses, and erosion rates in complex natural samples. By merging time-resolved velocimetry and interface tracking with real-rock imaging, we are able to explore the underlying relationship between fluid shear and mechanical erosion.

We apply this platform to investigate erosion dynamics in limestone under acid injection. Our experiments reveal a flow-

driven transition between two erosion regimes—transport-limited and reaction-limited—mediated by the mobility of CO_2 bubbles generated during the reaction. In the transport-limited regime, confined liquid films and bubble-induced acceleration enhance both dissolution and detachment. In the reaction-limited regime, bubble adhesion reduces reactive surface availability and weakens shear stress. We propose a theoretical model that predicts the critical Péclet number marking this transition and derive scaling laws that quantify chemical and mechanical erosion rates across regimes. This integrated approach provides new insights into the coupling between bubble dynamics, flow behavior, and erosion, and it opens a pathway for linking microscale erosion phenomena with large-scale reactive transport models used in geoscience and subsurface engineering.

2. Materials and methods

2.1 Fabrication of real-rock microfluidics

To investigate erosion dynamics in mineralogically diverse rocks under multiphase flow environments, we fabricate real-rock microfluidic chips using three representative carbonate samples with distinct mineral compositions: pure calcite, homogeneous limestone, and heterogeneous oolitic limestone (Table 1). The pure calcite sample (100% calcite) and the homogeneous limestone (93% calcite, 5% dolomite, and 2% other minerals by mass) exhibit relatively simple mineralogical structures (Fig. 1b). These two rocks are used to validate the accuracy and reproducibility of our experimental measurements (section 2.4). To explore the coupled processes of chemical dissolution and mechanical erosion in complex mineralogical systems, we select oolitic limestone—a common reservoir rock in oil and gas formations.[58,59] X-ray diffraction (XRD) analysis of the oolitic limestone reveals a composition of 73.8% calcite, 25.4% dolomite, and 0.5% wollastonite by mass (Fig. 1b). Further mineral mapping using a Tescan Integrated Mineral Analyzer (TIMA) indicates that dolomite, wollastonite, and clay minerals are embedded within a calcite matrix. The diameter of ooids ranges from 240 μm to 670 μm, which is smaller than the microfluidic channel length (3200 μm), ensuring that the embedded sample adequately captures the rock's intrinsic mineral heterogeneity. Previous studies have shown that dolomite dissolves approximately one order of magnitude slower than calcite,[60,61] making the residual dolomite more susceptible to mechanical erosion after calcite dissolution.

Microfluidic devices are fabricated from these rock samples using the following procedure. First, a rectangular carbonate block is cut into ~500 μm thick slices (3.2 mm × 3.2 mm) using a thin-sectioning saw (FEMA Fcut 230). The slices are subsequently polished to a target thickness of ~200 μm using a precision polishing machine (FEMA Fpol 252A Pro). Each polished slice is then divided into two halves with a diamond cutter to simulate a fractured geometry. These two half-slices (dimensions: 3.2 mm × 1.6 mm × 200 μm) are mounted on a polydimethylsiloxane (PDMS) flow cell, molded from a silicon wafer master. A gap of ~200 μm is maintained between the rock slices to represent a fracture aperture. To seal the device, a glass slide pre-coated with a thin PDMS film is bonded to the PDMS chamber using oxygen plasma treatment (Harrick Plasma, PDC-002). The entire chip is then clamped using a custom-built holder equipped with four bolts to maintain a secure and uniform seal. The device thus forms a quasi-two-dimensional (quasi-2D) microfluidic chip, with a 200 μm-thick rock sample embedded in the central flow chamber (Fig. 1a). This configuration enables direct optical observation of mineral-specific dissolution and erosion under controlled hydrodynamic and chemical conditions using soft lithography-based visualization. X-ray microscopy revealed that the matrix porosity of the oolitic limestone is extremely low (0.15%) with an average pore size of ~8 μm (Fig. S1). To evaluate the permeability of the matrix, we applied a pore-throat bundle approximation based on the Kozeny–Carman relationship, incorporating pore-throat scaling and tortuosity.[62] This analysis indicates that the upper bound of the matrix permeability is on the order of 10^{-14} m^2 (Fig. S1). In contrast, the fracture permeability, estimated from a 200 μm aperture channel, is ~3×10^{-9} m^2. Therefore, the fracture is 5 orders of magnitude more permeable than the matrix, indicating that flow and transport are

Table 1 Summary of erosion experiments under three rock types and flow velocity conditions

Sample description	exp. #	Q (μL min^{-1})	Pe = $Q/(D_m h_0)$	$\langle h_0 \rangle$ (μm)	$\langle b_0 \rangle$ (μm)
Heterogeneous oolitic limestone[a]	1, 10, 19	0.24	10	204	206
	2, 11, 20	0.43	18	201	188
	3, 12, 21	0.76	32	204	208
	4, 13, 22	1.35	55	205	214
	5, 14, 23	2.40	99	204	197
	6, 15, 24	4.27	171	204	222
	7, 16, 25	7.59	304	206	224
	8, 17, 26	13.50	546	207	192
	9, 18, 27	24.00	990	205	188
Calcite post[b]	28	2.4	100	201	—
Homogeneous limestone[b]	29[b]	24	1000	205	208

[a] exp #10–18 and #19–27 represent the repeated experiments with the same flow rate as exp #1–9. [b] exp #28 and exp #29 with simple mineral distributions are used to evaluate the accuracy of our quantification method. The average initial thickness h_0 = 204 μm, and the average initial aperture b_0 = 204 μm.

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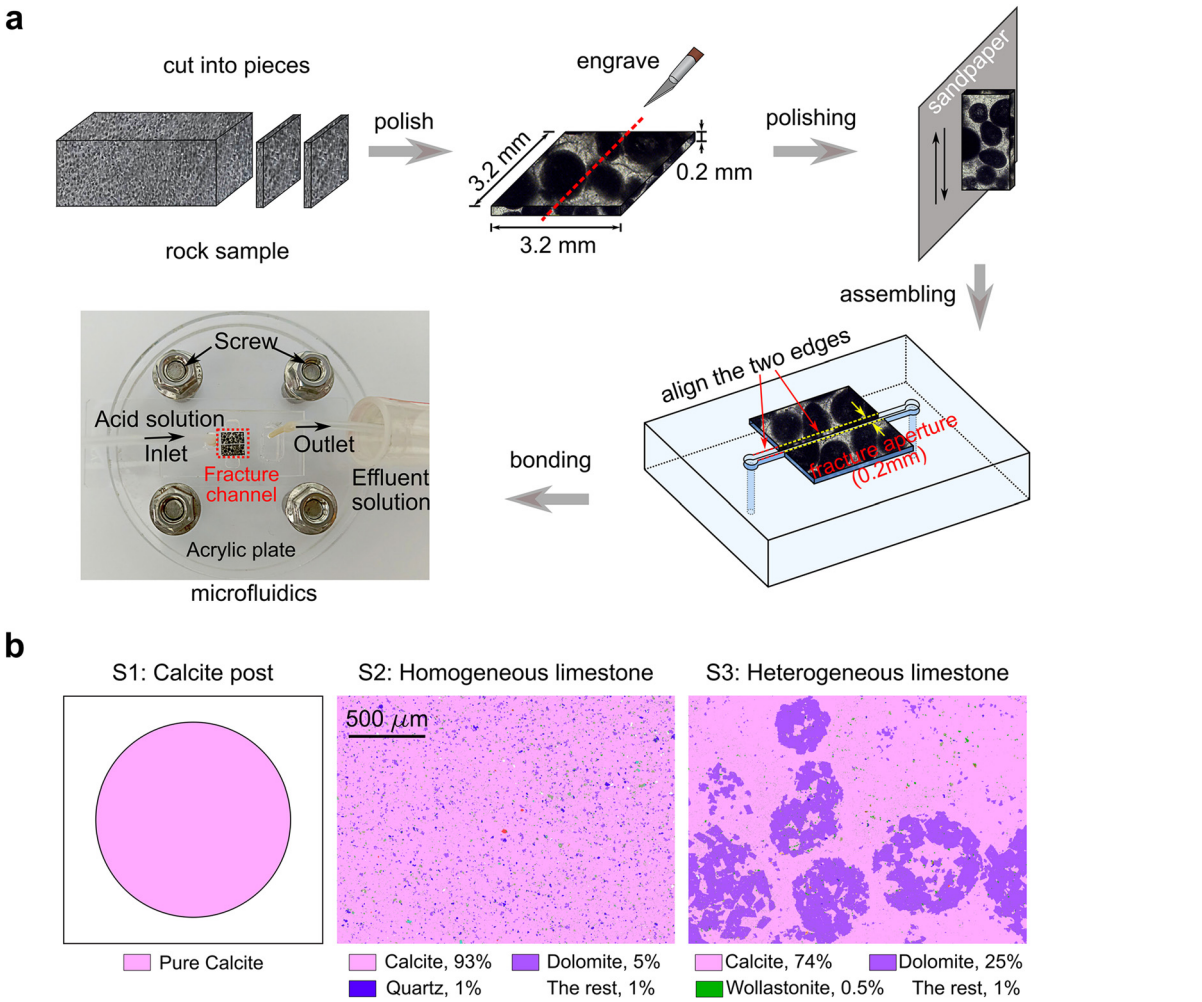


Fig. 1 (a) Fabrication of the real-rock microfluidic device. (b) Characterization of rock mineral composition by X-ray diffraction (XRD) and mineral distribution with a Tescan Integrated Mineral Analyzer (TIMA).

mainly controlled by the fracture, and matrix infiltration can be neglected in our experiments.

2.2 Flow-visualization system

To observe erosion dynamics and measure flow fields in rock fractures, we develop an integrated platform combining inverted fluorescence microscopy below the sample stage with microparticle image velocimetry (micro-PIV) above it, as illustrated in Fig. 2d and detailed in Fig. S2. This setup enables real-time observation of acid flow, rock erosion, and multiphase interactions at a controlled temperature of 25 °C. Acid solutions (pH = 1.0), dyed with a Super Fluor 488 to ensure reaction-induced multiphase flow are injected *via* a syringe pump (Harvard Apparatus PHD Ultra). Excited by blue light, the dye emitted green fluorescence, distinguishing dynamic acid–CO_2–rock interfaces (Fig. 2a). A CMOS camera (Axiocam 305) records reaction-induced erosion and bubble generation at 1.3 μm per pixel resolution (Fig. 2b). Experiments last 30 minutes to 2 days, depending on injection rates (Table 1). Fluorescent particles of different sizes were used for qualitative streamline observation and quantitative flow field measurement. The use of 2 μm particles allows a relatively smaller interrogation window size, which improves the spatial resolution and accuracy of the flow field calculation. Moreover, for qualitative visualization of fluid streamlines, the larger 5 μm particles are used to enhance the width and visibility of the streamline. Time-lapse imaging tracks the fluid streamline evolution: an acid mixed with 5 μm fluorescent particles is injected and excited by blue light. At Pe ≈ 1000 (mean velocity $v \approx 10$ mm s^{-1}) and 0.3 s exposure, particles traced streaklines are comparable to the channel length (~3.2 mm), visualizing flow deformation (Fig. 2a). For micro-PIV (LaVision) quantification (Fig. 2c), the fracture is first saturated with deionized (DI) water, followed by an acid (pH 1.0) seeded with 2 μm fluorescent polystyrene particles (RigorBiotech). The mass concentration of the particles was maintained at approximately 1 mg mL^{-1}, corresponding to a nominal volume fraction of 9.5×10^{-4}. This concentration ensured a seeding density of 6–8 particles per 16 × 16 pixel interrogation window, which is adequate for reliable velocity

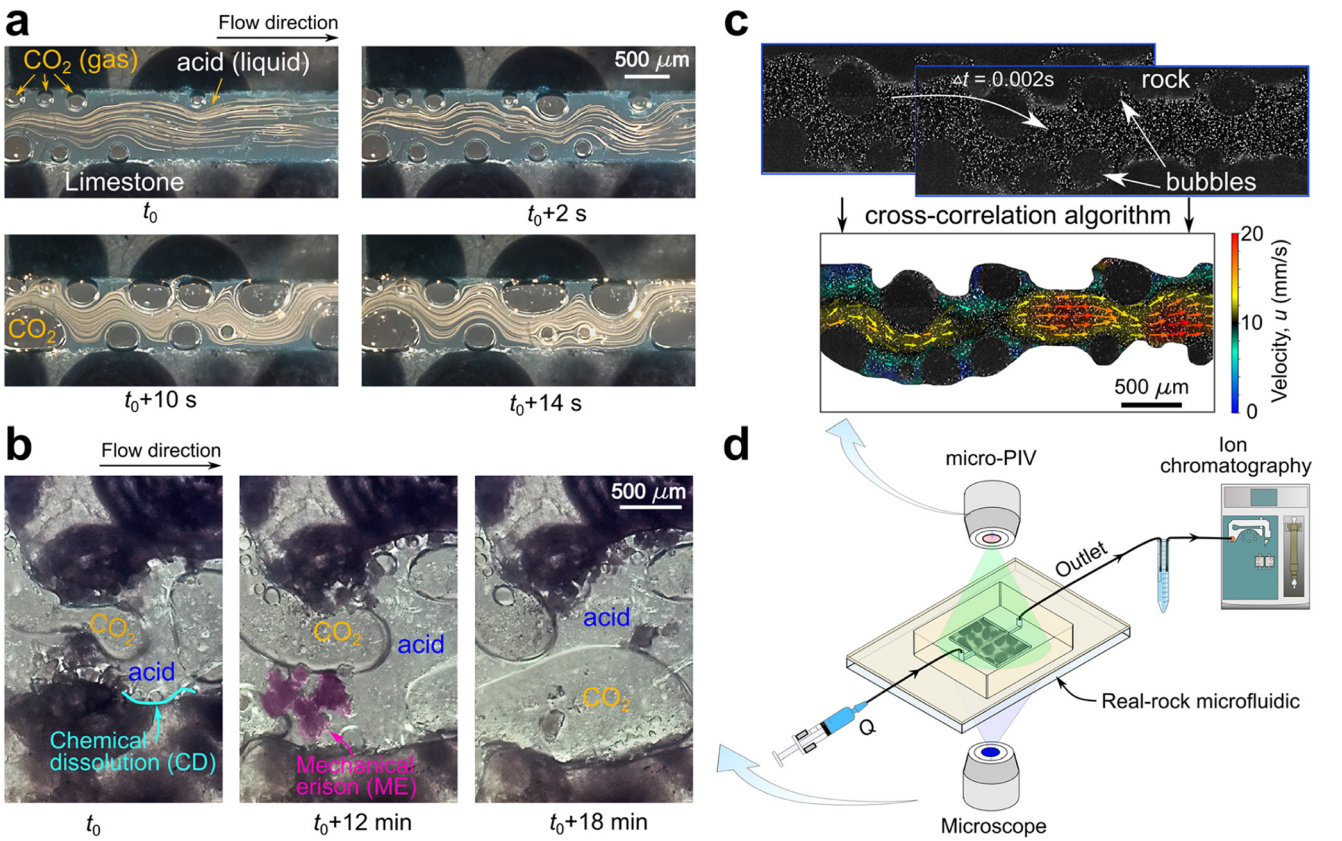


Fig. 2 Experimental platform for visualizing flow-erosion processes in real-rock microfluidics. (a) Time-lapse images showing streamline distortion caused by CO_2 bubbles, increasing flow tortuosity. (b) Microscopy captures calcite dissolution and dolomite detachment (purple) at the rock surface. (c) Micro-PIV visualizes flow velocity fields in the multiphase system. (d) Schematic of the platform integrating microscopy, micro-PIV, and ion chromatography. The setup enables simultaneous observation of flow paths, interface dynamics, and mineral detachment, with effluent cation concentrations (Ca^{2+}, Mg^{2+}) measured at the outlet.

measurements. With an image resolution of 3.5 μm per pixel, an interrogation window size of 16 × 16 pixels with a 50% overlap yields a vector spacing of 28 μm and a spatial resolution of 56 μm. Frame pairs were acquired at a frequency of 10 Hz, with an exposure time of 2.5 ms per frame, synchronized with the laser pulse duration. The time separation (Δt) between the two pulses in a pair was adjustable. The displacement (Δx) of tracer particles over Δt was tracked within each interrogation window using commercial software Davis 10.0 *via* cross-correlation analysis, thereby obtaining the full flow field and the shear rate distribution across the fracture (Fig. 2c).

Our multi-modal optical imaging techniques provide critical insights into the role of multiphase flow in heterogeneous rock erosion. The chemical reaction governing the acid-driven dissolution of the carbonate rock—involving calcite dissolution and CO_2 generation—is presented as follows:

$$CaCO_3(s) + 2HCl(aq) \rightarrow CaCl_2(aq) + CO_2(g) + H_2O(l)$$

Microparticle tracking revealed streamline distortion from reaction-generated CO_2 bubbles (Fig. 2a). For example, the initially uniform flow streamlines are observed at t_0, and streamlines became increasingly tortuous as bubbles grew and coalesced (t_0 + 2 s, t_0 + 10 s, t_0 + 14 s), narrowing channels and altering flow pathways. The influence of bubbles on fluid pathways under different flow rates was systematically investigated across different flow rates, as shown in Fig. 7. Micro-PIV non-invasively quantified evolving velocity fields (Fig. 2c), linking bubble mobility to erosion dynamics and fracture morphology changes, as further explored in Fig. 5. High-resolution microscopy capture fluid–rock interface evolution: after ~12 minutes, calcite dissolution weakened bonds between dolomite and clay minerals, leading to detachment and mobilization of non-cohesive particles by fluid drag force at t_0 + 18 minutes, directly evidencing mechanical erosion (Fig. 2b). These observations elucidate how gas generation and fluid mechanics interplay to drive erosion heterogeneity, with implications for microfluidic devices and geological applications in fractured systems.

2.3 Workflow to quantify chemical dissolution and mechanical erosion

To quantify chemical dissolution and mechanical erosion rates in rock fractures, we develop a workflow based on our

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experimental platform (Fig. 3a). Total erosion is assessed using high-resolution microscopy images (Fig. 3b and S3). Original fluorescence images are processed in ImageJ to distinguish bubbles and liquid regions from rock based on color differences. The images are then segmented into rock and channel regions, allowing extraction of rock-fracture interface profiles at initial time t_0 and $t_0 + \Delta t$. The total eroded volume is calculated as $\Delta V_{TOT} = V_f(t_0 + \Delta t) - V_f(t_0)$, where the fracture volume $V_f = S_f h_0$, with S_f being the fracture channel area from 2D images and $h_0 \approx 0.2$ mm being the sample thickness. This volume is derived from post-processed images at t_0 and $t_0 + \Delta t$ (highlighted in blue, Fig. 3b). To maintain a constant injected acid volume of 100 μL per interval, Δt is adjusted according to the Péclet number (Pe): approximately 416.7 minutes for Pe = 10, 41.7 minutes for Pe = 100, and 4.2 minutes for Pe = 1000. This ensured consistent conditions for comparing erosion rates across different flow regimes.

Chemical dissolution is quantified using ion chromatography (Metrohm Eco IC) to analyze effluent solutions for ionic concentrations (Fig. 3c and S4). Calibration involves measuring conductivity curves of standard solutions containing Na^+, K^+, Mg^{2+}, and Ca^{2+} at concentrations of 2, 4, 6, 8, and 10 mg L^{-1}. Peak positions in the conductivity curves identify specific ions, while peak areas correlate with concentrations, as exemplified by the 10 mg L^{-1} standard (Fig. 3c). Effluent from erosion experiments is analyzed under identical conditions. For instance, at Pe = 100, ion concentrations are determined as 0.12 mg L^{-1} (Na^+), 1.2 mg L^{-1} (K^+), 20.78 mg L^{-1} (Ca^{2+}), and 0.22 mg L^{-1} (Mg^{2+}) by comparing peak areas. The chemically dissolved volume ΔV_{CD} is calculated from Ca^{2+} and Mg^{2+} concentrations, reflecting dissolution of calcite and dolomite in heterogeneous limestone reacting with hydrochloric acid. Notably, Ca^{2+} was the dominant ion, with Mg^{2+} concentrations two orders of magnitude lower, indicating negligible dolomite dissolution during the experiments. The effluent solution was collected at

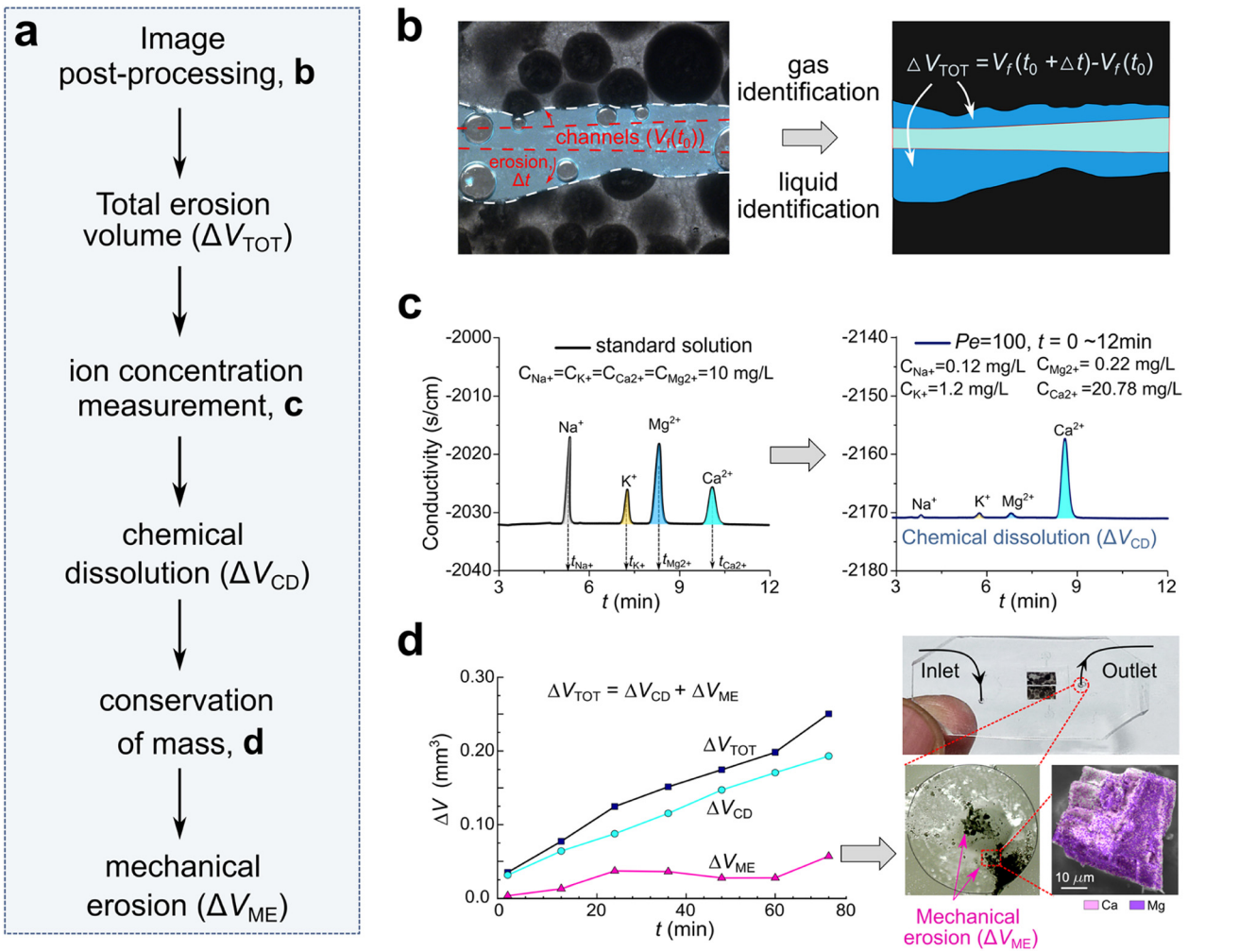


Fig. 3 (a) Workflow to quantify the chemical dissolution and mechanical erosion rate. (b) The volumes of total erosion (ΔV_{TOT}) are determined from binarization and comparison of microscope images captured during a time interval. (c) The chemical dissolution volume (ΔV_{CD}) is measured by the ion chromatograph calibrated with the standard solution. (d) The mechanical eroded volume (ΔV_{ME}) is calculated as $\Delta V_{TOT} - \Delta V_{CD}$ according to the mass conservation law, with mineral distribution analysis identifying dolomite particles as the mechanically eroded minerals migrating to the outlet.

the outlet after the reaction with the rock samples throughout the fracture. Each sample was collected in a fixed volume of 100 μL for measuring the concentration of dissolved species, which represents the chemical dissolution volume of the entire fracture channel.

Mechanical erosion is derived by subtracting chemical dissolution from total erosion, leveraging mass conservation and assuming negligible pore volume in the limestone matrix compared to the eroded rock volume. Thus, the mechanical erosion volume is $\Delta V_{ME} = \Delta V_{TOT} - \Delta V_{CD}$ (Fig. 3d). The microscopy, coupled with a camera, can capture the entire 3.2 mm length of the fracture. Thus, the image-based method captures the total erosion volume of the entire fracture–rock system, while ion chromatography measures the chemical dissolution volume of the whole system. Thus, the mechanical erosion volume (ΔV_{TOT}) calculated from these values represents the erosion volume of the full fracture–rock system. Microscopy captures the dynamic transport of detached insoluble mineral particles under fluid shear stress (Fig. 2b), providing visual evidence of mechanical erosion processes. Particles collected at the outlet are characterized using scanning electron microscopy (SEM) and X-ray diffraction (XRD) to determine the mineral composition and crystalline structure (Fig. S5). Results show that eroded particles primarily consisted of dolomite ($CaMg(CO_3)_2$), confirming selective mechanical erosion of the slower-reacting dolomite component (Fig. 3d). This integrated platform achieves high spatiotemporal resolution imaging (1.3 μm per pixel, 1000 fps) of erosion dynamics in multiphase flow environments, while enabling separate quantification of chemical and mechanical erosion rates—capabilities not previously realized in similar systems. By decoupling these processes, the workflow provides deeper insights into erosion mechanisms in fractured rocks, with applications in microfluidics, geochemistry, and enhanced oil recovery. The approach enhances reproducibility and accuracy in quantifying multiphase interactions, addressing limitations in traditional methods that often conflate chemical and mechanical contributions.

2.4 Validation of the workflow

To validate the accuracy of our method, we perform microfluidic experiments on two rock types with simple and well-characterized mineral compositions: pure calcite and homogeneous limestone. These samples are chosen due to their minimal mineral heterogeneity, which allows for clear validation of chemical *versus* mechanical erosion measurements. In the first experiment, a well-controlled test using a pure-calcite micro-post is performed (Fig. 4a). For this system, the chemically dissolved volume (ΔV_{CD}) should be equal to the total eroded volume (ΔV_{TOT}), as calcite undergoes uniform dissolution without producing mechanically detached debris. ΔV_{TOT} is determined from time-lapse images, while ΔV_{CD} is estimated using chromatographic analysis. The discrepancy between these two independent measurements is within 2.8% (Fig. 4c), confirming the reliability of our quantification approach under ideal conditions.

To further evaluate the generalizability of our method in more complex but still controlled systems, we apply it to a microfluidic model composed of calcite-rich limestone (Fig. 4b), which has a relatively homogeneous composition of 93% calcite, 5% dolomite, and 2% quartz and other minerals by volume. In such a system, chemical dissolution is expected to dominate, with mechanical erosion accounting for less than 7% of the total volume loss. Our measurements show that the mechanically eroded mass contributed approximately 4.8% of the total eroded volume (Fig. 4d). This value is higher than the volume fraction of low-reactivity minerals (2%), indicating that even minerals like dolomite with moderate reactivity may contribute to mechanical detachment. These results demonstrate that the workflow can effectively distinguish between chemical and mechanical erosion, even in rocks with minor mineralogical complexity. Collectively, these validation experiments confirm the accuracy and robustness of our integrated platform and workflow in quantifying erosion dynamics under multiphase flow conditions.

3. Results and discussion

3.1 Observation of erosion regimes

Using the above-described microfluidic platform, we systematically investigate erosion dynamics in limestone fractures under varying hydrodynamic conditions. Flow rates are varied from 0.24 to 24 μL min^{-1}, corresponding to Péclet numbers ($Pe = vb_0/D_m$, where $v = Q/(h_0 b_0)$) ranging from 10 to 1000—spanning two orders of magnitude relevant to both fractured media and microfluidic systems. Prior to acid injection, the fracture channel is saturated with deionized water. Hydrochloric acid (pH = 1) is then injected at controlled flow rates. The acid–rock reaction generated CO_2, forming dynamic fluid–fluid interfaces characteristic of multiphase flow. This configuration is representative of various geological and engineering systems such as karstification, hydrate recovery, and acid stimulation. The evolving erosion morphologies across all Pe conditions are presented in Fig. S6. For each Pe condition, triplicate experiments were conducted to ensure reproducibility (see Fig. S7), totaling 27 experiments (Table 1).

Fig. 5a illustrates how erosion patterns transition with an increasing flow rate. At low Pe (*e.g.*, Pe = 10 and 32), significant erosion is concentrated near the inlet, while downstream regions remain largely intact, resulting in a converging channel geometry (Fig. 5a, rows 3–4). This behavior resembles the classical transport-limited regime observed under single-phase flow conditions, where solute transport constrains erosion. However, in our multiphase system, additional features emerge due to the generation and immobilization of CO_2 bubbles. As shown in Fig. 2b, these bubbles occupy the central void, confining the flow to thin liquid films along the fracture surfaces. These films enable localized transport of reactive fluid and mineral debris

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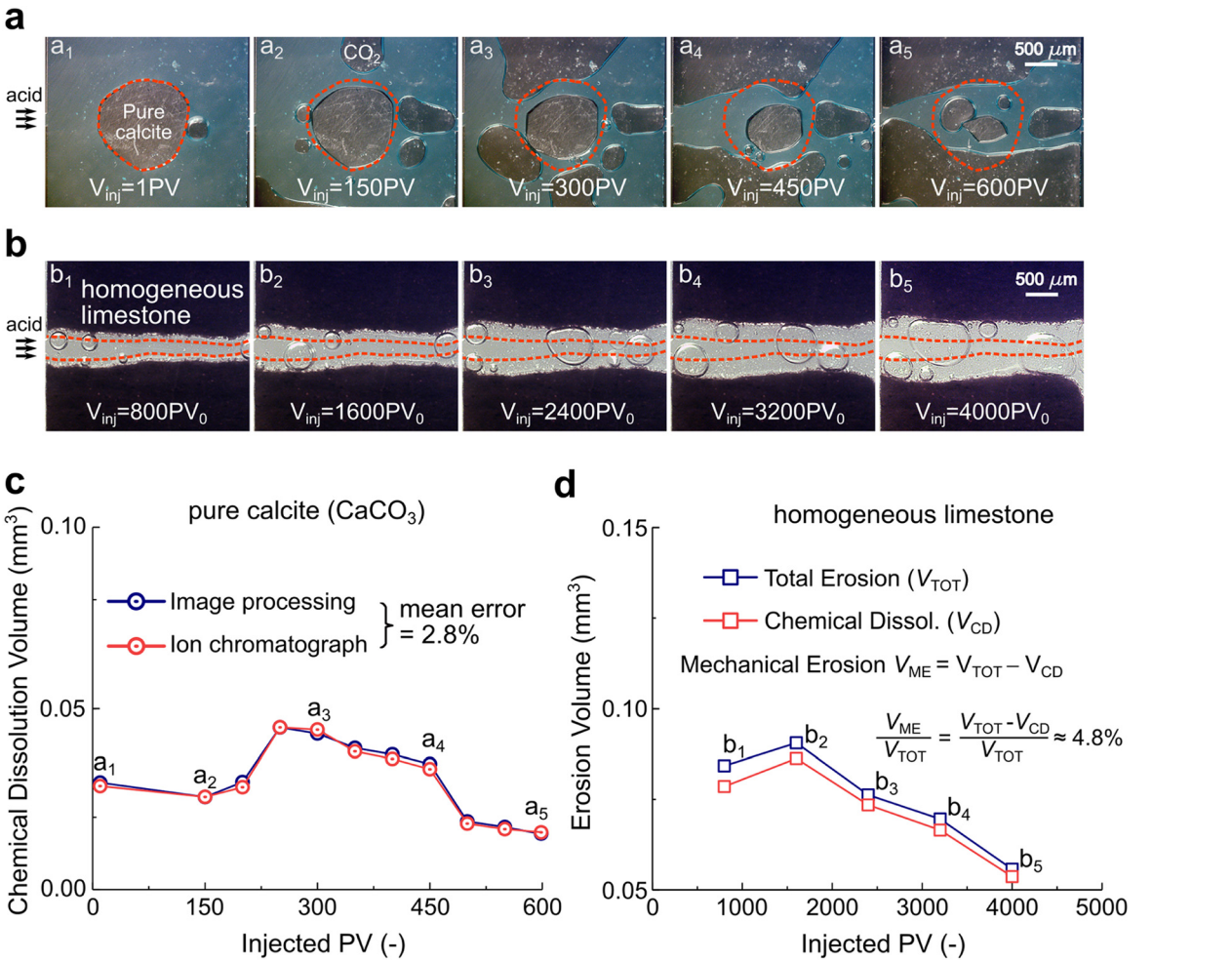


Fig. 4 Validation of quantification workflow using calcite and homogeneous limestone. (a_1–a_5) Geometry evolution of calcite post under acid injection (pH = 1, Q = 24 μL min^{-1}, Pe = 1000). (b_1–b_5) Geometry evolution of homogeneous limestone channel under acid injection (pH = 1, Q = 24 μL min^{-1}, Pe = 1000). (c) ΔV_{CD} and ΔV_{TOT} show a ~2.8% error, confirming that dissolution is dominant in pure calcite. (d) In limestone, ~4.8% of ΔV_{TOT} is attributed to mechanical erosion (ΔV_{ME}), consistent with the volume fraction of inert minerals (~7%). The workflow reliably distinguishes chemical and mechanical erosion components.

(Fig. 5b). Although the rock surface is exposed to acidic solution throughout the channel (the red curve in Fig. 2b), the overall erosion remains transport-limited due to restricted solute mobility. Compared with single-phase transport-limited regimes, bubble presence exacerbates inlet-localized dissolution, thereby intensifying the convergence of the channel geometry (Fig. 5b and Movie S1).

At higher Pe (Pe ≥ 100), erosion becomes more uniformly distributed (Fig. 5a, rows 1–2), indicating improved fluid access throughout the channel. Under such conditions, erosion would typically be expected to fall into the reaction-limited regime, as the Damköhler number (Da = k/v) ranges from 10^{-4} to 10^{-2}. However, our results show that this classical framework does not fully describe the erosion behavior in a multiphase system. At high flow rates, enhanced shear forces increase bubble mobility, reducing bubble residence time and minimizing their inhibitory effect on erosion. Consequently, more reactive calcite dissolves preferentially, exposing less reactive minerals like ooids (Fig. 5a, rows 1–2). This implies that erosion patterns are increasingly governed by mineral heterogeneity rather than bubble blockage. Nonetheless, before detachment, CO_2 bubbles adhere to the rock surface (Fig. 5c and Movie S2), temporarily reducing the reactive surface area. Compared to single-phase systems, these adhered bubbles inhibit both chemical dissolution and particle detachment. Thus, while this regime aligns with the reaction-limited classification, its behavior under multiphase flow is distinctly different. The unique features of this multiphase reaction-limited regime are further quantified *via* erosion rate (Fig. 6) and velocity field measurements (Fig. 7) in the next subsections.

3.2 Bubble dynamics controlling erosion

To elucidate the mechanism driving the erosion regime transition observed in Fig. 5, we analyze the dynamics of CO_2 bubbles and their influence on eroded volumes—total erosion (ΔV_{TOT}), chemical dissolution (ΔV_{CD}), and mechanical erosion (ΔV_{ME})—under varying hydrodynamic conditions (Fig. 6). At a representative injected volume of V_{inj} = 3200

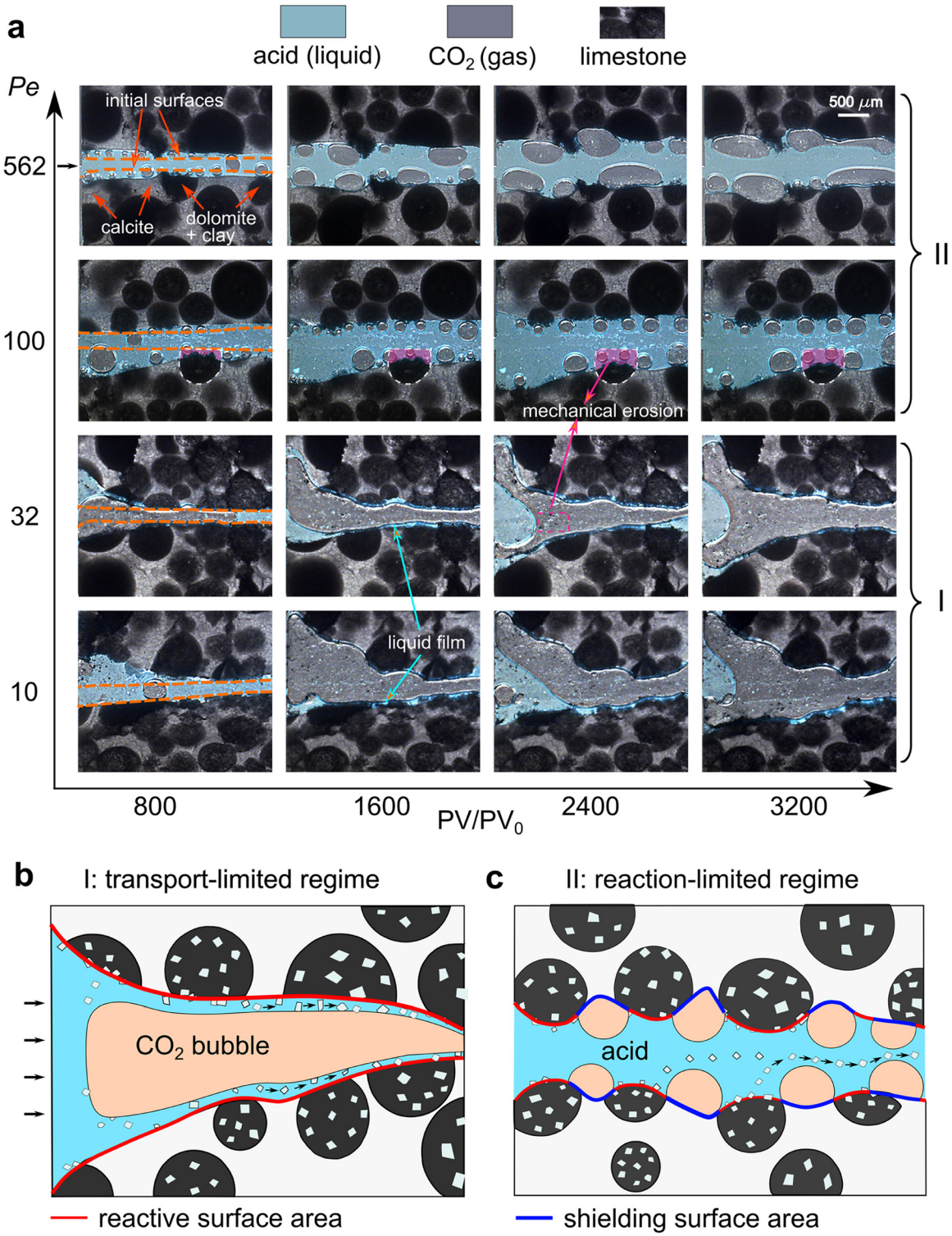


Fig. 5 Erosion morphologies for different injection rates. (a) Erosion process for different injected rates (bottom to top: Pe = 10, 32, 100, 562) and injected volumes (left to right: injected volume/initial pore volume = 800, 1600, 2400, 3200). (b) Schematic diagram of the transport-limited regime in the multiphase environment. The liquid films serve as the flow pathway for the acid and the detached minerals due to the bubbles blocking the center of the channel. (c) Schematic diagram of the reaction-limited regime in the presence of gas bubbles. Before detachment from the rock surfaces, the CO_2 bubbles attached to the surfaces shield the acid–rock contact area, and the erosion has been suppressed.

PV_0, these eroded volumes display a nonmonotonic dependence on the Péclet number (Pe), indicating the existence of a critical flow threshold (Pe_{cr}) that separates two distinct erosion regimes: transport-limited and reaction-limited (Fig. 6d–f). For $Pe < Pe_{cr}$, CO_2 bubbles nucleate and merge transversely along the rock surface, eventually forming

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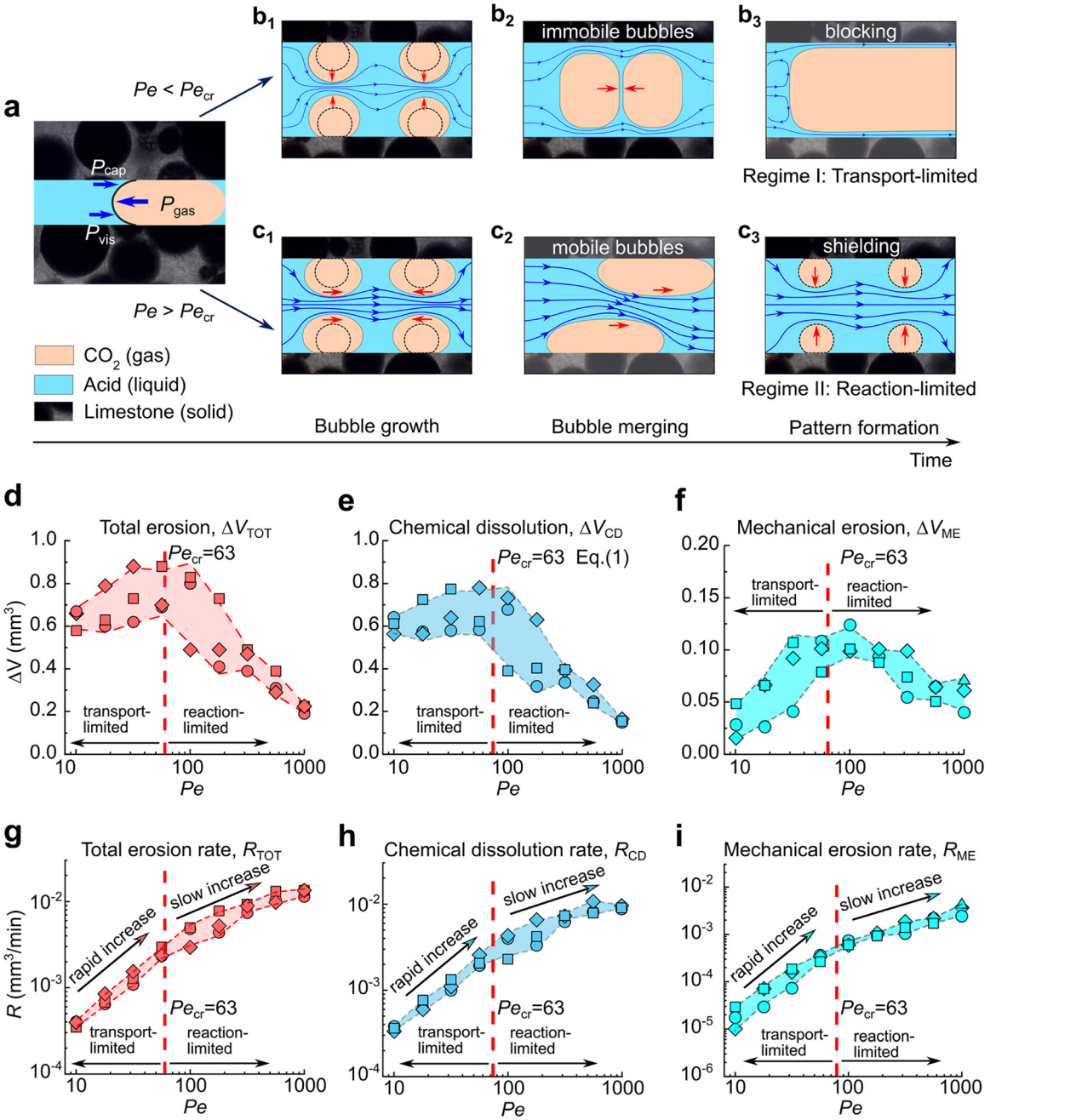


Fig. 6 Quantification of erosion rates. (a–c) Formation of different erosion regimes controlled by the interplay between bubble mobility and hydrodynamics. At low flow rates, low hydrodynamic conditions result in the growth and transverse merging of bubbles on the upper and lower walls (b_1), causing the CO_2 bubbles to occupy the center of the fracture channel (b_2). The fluid driving force is inadequate to propel the large bubble out, confining the acid flowing as a liquid film between the CO_2 bubbles and the wall (b_3). At high flow rates, the increased fluid pressure promotes the detachment of growing bubbles adhered to the rock surfaces (c_2). Before detachment, these bubbles impede the contact between the acid and the rock surface (c_1). Once detached, a new cycle of acid–rock reaction occurs, producing nascent bubbles that shield the rock surface and reduce the effective reactive area (c_3). (d–f) Total erosion volume (ΔV_{TOT}), chemical dissolution (ΔV_{CD}), and mechanical erosion volume (ΔV_{ME}) as a function of Pe, where the erosion volumes are quantified within a fixed acid injection volume (V_{inj} = 3200 PV_0). (g–i) The evolution characteristics of the total erosion rate (R_{TOT}), chemical dissolution rate (R_{CD}), and mechanical erosion rate (R_{ME}) with the Pe number. The points represent the data from repeated experiments, and the bars indicate the associated uncertainties. The erosion experiment was repeated three times for each Pe number.

longitudinally aligned bubbles that remain trapped within the fracture due to an insufficient pressure gradient (Fig. 6b_1–b_3). These bubbles obstruct the flow pathways and confine the reactive fluid to thin liquid films adjacent to the rock surface. Particle tracking experiment confirms that fluid transport occurs predominantly through these films (Movie S3). In this transport-limited regime, increased flow rate enhances the convective delivery of acid and increases shear

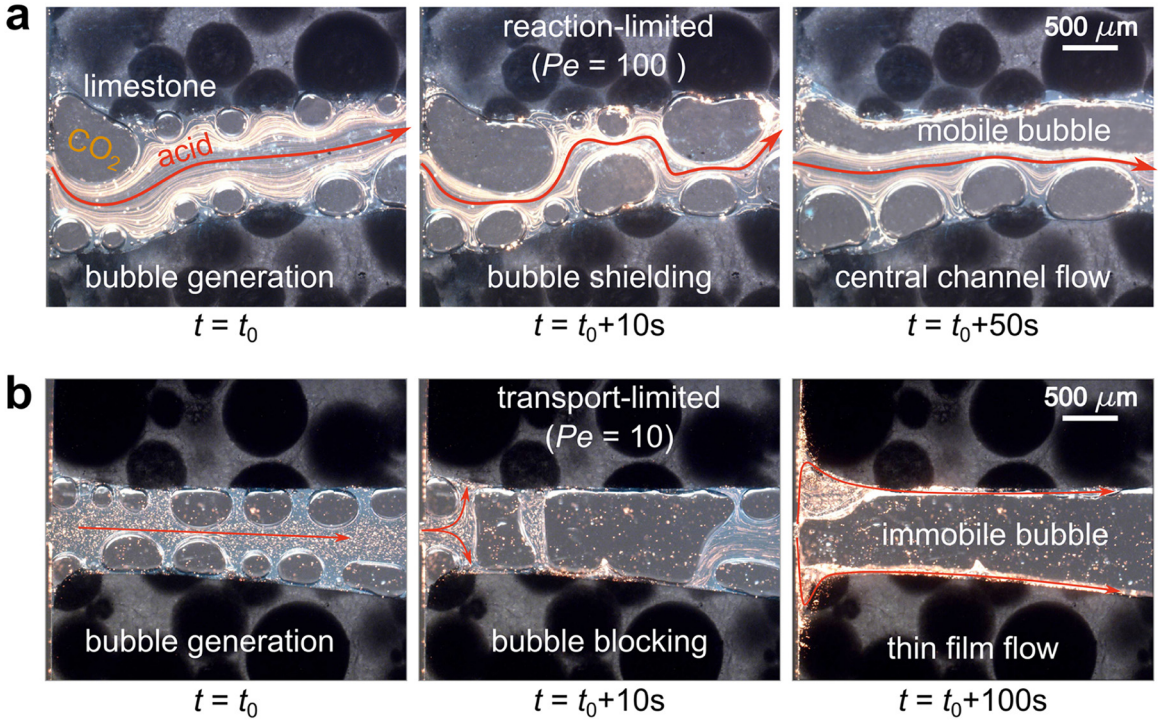


Fig. 7 The streamline of the reactive fluid characterized by time-lapse photography. (a) Bubbles growing on the fracture walls shield the rock surface, causing the acid to flow rapidly through the fracture channel but limiting its reaction with the rock samples. (b) Immobile bubbles force the reactive fluid to migrate through the thin liquid film between the bubbles and the rock surfaces.

forces within the confined films, promoting both dissolution and particle detachment. As a result, ΔV_{CD} and ΔV_{ME} increase with Pe within this regime.

When Pe exceeds Pe_{cr}, the pressure gradient becomes sufficient to mobilize and displace gas bubbles out of the fracture network (Fig. 6c$_2$). Although residual bubbles adsorbed on the rock surface persist (Fig. 6c$_1$ and c$_3$), they no longer block the primary flow channel, allowing the reactive fluid to penetrate rapidly into the fracture (Movie S4). However, due to the reduced residence time of the fluid, the extent of acid–rock reaction diminishes. As a result, ΔV_{TOT}, ΔV_{CD}, and ΔV_{ME} decrease with further increases in Pe, reflecting a reaction-limited regime. Notably, erosion efficiency reaches a maximum at intermediate Pe values, highlighting Pe_{cr} as the transition point where flow conditions best balance transport enhancement and reaction time. Direct visual observation of detached minerals at the outlet further supports this conclusion: mechanical erosion volume (ΔV_{ME}) is highest at Pe $\approx$ 100 and lower at both Pe = 10 and Pe = 1000 (Fig. S5). This nonmonotonic trend underscores the need to understand and predict Pe_{cr} for multiphase flow-driven erosion systems.

To theoretically predict Pe_{cr}, we consider the critical condition when CO_2 bubbles begin to move within the fracture channel. At this point, the fluid flow is governed by a force balance between driving forces—viscous pressure (P_{vis}) and capillary pressure (P_{cap})—and the resistance from the gas bubble pressure (P_{gas}) (Fig. 6a). The viscous pressure, derived from Darcy's law, is $P_{vis} = \mu Q_{in}\delta_x/(KA_{in})$, where μ is the fluid viscosity, δ_x is the characteristic length, A_{in} is the cross-sectional area, and K is the effective permeability of the channel, estimated as $K = \alpha b_0^2/12$ where α is an attenuation factor that accounts for the wall roughness, interfacial friction, and other sources of flow resistance (Table S1). The value of α was calibrated in our previous work on a smooth limestone fracture channel under comparable conditions,[30] namely a multiphase dissolution environment (pH = 1) and Pe ranging from 10 to 1000. In the present system, the fracture aperture b_0 was adopted as the characteristic length scale for permeability estimation because the channel depth h_0 and fracture aperture were of the same order of magnitude ($h_0 = b_0 \approx 200\,\mu m$). It is worth noting that if the initial aperture exceeds the channel thickness ($b_0 > h_0$), the smaller dimension (h_0) would instead control flow resistance, leading to a permeability scaling of $h_0^2/12$. The capillary pressure, from the Young–Laplace equation, is $P_{cap} = \beta\sigma/b_0$, with σ as the surface tension, b_0 as the fracture aperture, and β as the capillary coefficient (Table S1). The gas bubble pressure is estimated from the ideal gas law as $P_{gas} = 2k\Delta t\rho CO_2 RT/(M_{CO_2} b_0)$, where k is the CO_2 production rate, ρ_{CO_2} is the CO_2 density, R is the gas constant, and T is the temperature. Substituting these terms into the force balance yields a theoretical expression for the critical Péclet number:

$$Pe_{cr} = \frac{K}{\mu L D_m}\left(\frac{2k\rho_{CO_2}RT}{M_{CO_2}} - \beta\sigma\right) \quad (1)$$

Details of the derivation are provided in Text S1 and our previous work.[30] All parameters have clear physical interpretations

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and are listed in Table S1. The resulting theoretical value of Pe_{cr} = 63 (the red dashed line in Fig. 6) agrees well with experimental observations, validating this threshold as the control point for erosion regime transition.

To further quantify erosion behavior across these regimes, we consider erosion rates: the total erosion rate ($R_{TOT} = \Delta V_{TOT}/\Delta t$), chemical dissolution rate ($R_{CD} = \Delta V_{CD}/\Delta t$), and mechanical erosion rate ($R_{ME} = \Delta V_{ME}/\Delta t$), where $\Delta t = V_{inj}/Q$ is the reaction time based on the injection volume (V_{inj}) and flow rate (Q). By normalizing erosion to the reaction time, these rates focus on the effect of Pe on erosion dynamics. As shown in Fig. 6g–i, all three rates increase with Pe, indicating that higher flow velocities enhance both solute transport and shear stress on mineral surfaces. However, the rates grow more rapidly in the transport-limited regime and plateau or increase slowly in the reaction-limited regime. This transition reflects the changing role of gas bubbles in erosion: in the transport-limited regime, stagnant bubbles constrict flow to thin films, making solute transport and shear stress the dominant controls on erosion. Here, increasing Pe substantially improves both transport capacity and mechanical mobilization of detached particles. In contrast, in the reaction-limited regime, although higher Pe facilitates bubble removal, residual surface bubbles—especially those trapped in dissolution pits—persist and shield the reactive surface (Fig. 5a). This "bubble shielding effect" limits the reactive rock area, thereby reducing the overall erosion rate. The pore-scale mechanisms governing this shielding phenomenon will be detailed in the following section.

3.3 Quantification of the shielding effect

Erosion within fractured rocks—comprising both chemical dissolution and mechanical detachment—is governed by the transport of reactive fluids and the migration of undissolved minerals. As both processes are highly sensitive to flow velocity, we employ time-lapse photography and micro-PIV to quantify how CO_2 bubble dynamics influence flow patterns and erosion behavior. Time-lapse imaging reveals two distinct regimes, as shown in Fig. 5: a transport-limited regime, where CO_2 bubbles remain immobilized at the fracture center (Fig. 7b), and a reaction-limited regime, where bubble evolution dynamically reshapes the flow field (Fig. 7a). In the transport-limited regime (Fig. 7b), the injected acid flows as a thin film between trapped bubbles and the rock surface, intensifying local mass transfer. As the injection rate increases, this confined flow leads to a sharp increase in erosion rates (Fig. 6g–i), primarily driven by enhanced advection within the narrow film. In contrast, in the reaction-limited regime, bubbles grow, move, and coalesce, altering fluid pathways and mineral transport (Fig. 7a). Flow streamlines become distorted upon bubble formation at t_0, and streamlines become increasingly tortuous as bubbles grow and coalesce (t_0 + 10 s), narrowing channels and altering flow pathways. As bubbles continue to expand due to chemical reaction, they confine the acid to the central region of the flow, thereby hindering contact between the reactive fluid and the rock surface (t_0 + 50 s). This process persists until the driving force becomes sufficient to displace the bubbles from the fracture. While these evolving structures can accelerate fluid transport through channel constriction, they also hinder direct acid–rock contact, leading to a reduced chemical interaction. This dual role of bubbles introduces complex feedback between flow reorganization and erosion progression.

To quantify the shielding effect introduced by CO_2 bubbles, we analyze the evolution of velocity fields and shear rates from micro-PIV measurements (Fig. 8 and Movie S5). Initially (Fig. 8a), flow is uniform and the entire rock surface within the observed region (region 1) is exposed to the reactive fluid, meaning the reactive surface area (A_{react}) is maximal. As bubbles form and expand (Fig. 8b–c), flow paths become increasingly tortuous, as shown by shifts in the probability density function of the velocity (Fig. 8e). Bubble growth progressively masks portions of the rock surface, reducing A_{react} (Fig. 8f) and shielding the rock from direct chemical attack. Simultaneously, local shear rates ($\dot{\gamma}$) near the surface decreases, especially in areas adjacent to large bubbles (region 3 in Fig. 8c and region 4 in Fig. 8d). The resulting decline in shear force ($f_s = \mu\dot{\gamma}A_{react}$) further suppresses mechanical erosion. At later stages (Fig. 8d), when bubbles coalesce into larger structures that adhere to the rock surface, both A_{react} and $\dot{\gamma}$ reach their minimum values, indicating maximal shielding. This quantification highlights how bubble growth in the reaction-limited regime reduces erosion despite increasing flow velocity. Notably, however, this shielding is only partial. Higher injection rates still promote bubble detachment *via* elevated shear forces, re-exposing the rock surface to acid and sustaining erosion. Thus, the observed erosion dynamics result from the competition between bubble-induced shielding and shear-induced bubble removal. While previous studies clarified how CO_2 bubble formation limits dissolution at the mineral surface,[27,63] our work advances this understanding by integrating micro-PIV and time-lapse imaging to directly quantify how bubble accumulation not only reduces surface exposure, but also reorganizes local streamlines and shear stresses. This coupled perspective on multiphase transport and surface shielding defines a distinctive aspect of reaction-limited erosion dynamics in fractured systems and highlights the interplay between bubble-induced shielding and the flow pathway evolution.

3.4 Scaling relationship for erosion rate

To investigate how reaction-limited conditions govern erosion dynamics under multiphase flow, we perform a scaling analysis on experimentally quantified erosion rates. Each experiment is repeated three times (Table 1), and average values of the chemical dissolution rate (R_{CD}) and mechanical erosion rate (R_{ME}), along with associated error bars, are evaluated under various flow conditions. Their dependence on Pe is shown in Fig. 9. In confined systems, both R_{CD} and R_{ME} follow the scaling forms:[64–67]

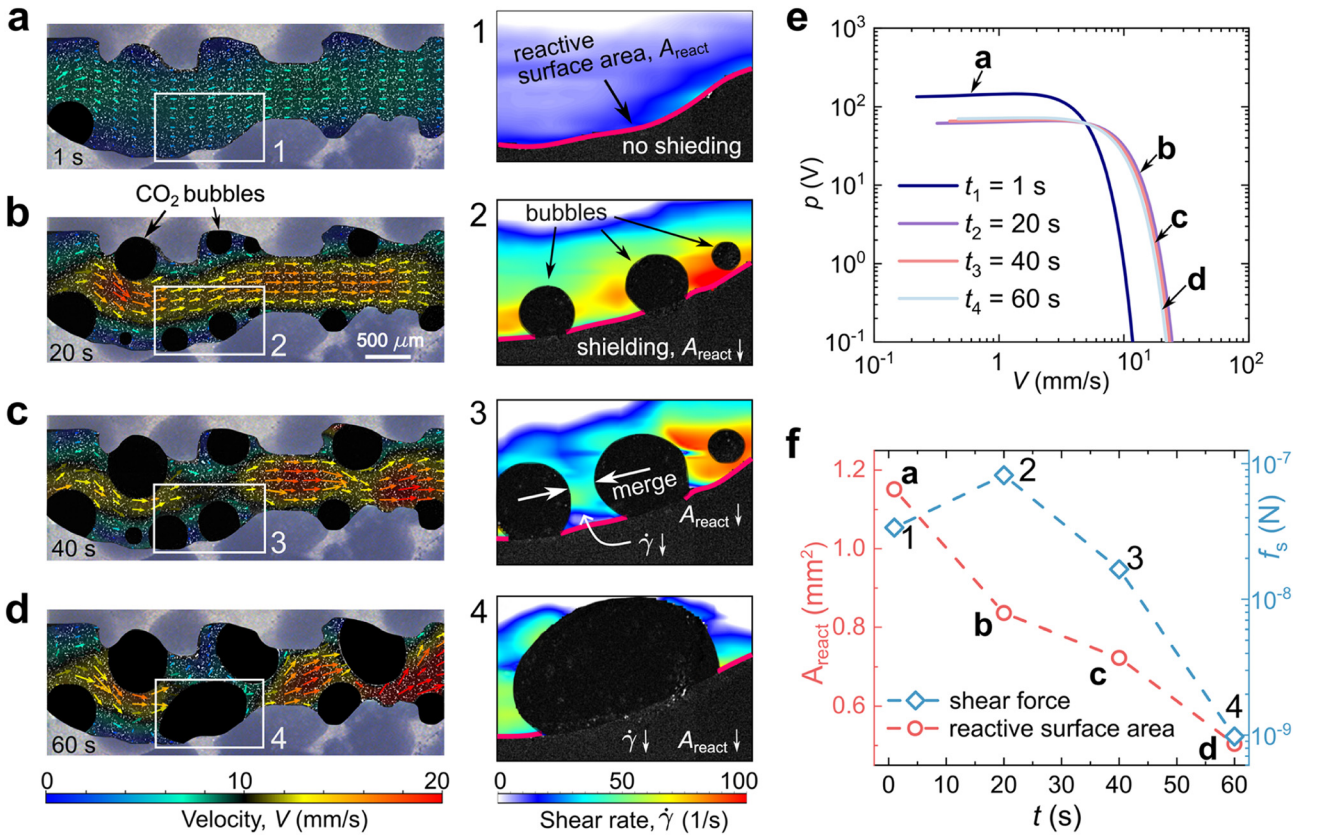


Fig. 8 The interaction between bubble dynamics and flow velocity fields quantified by micro-PIV measurements. (a–d) Velocity distribution within the channel at t = 1, 20, 40, and 60 s. The injected flow rate is 120 μL min^{-1} (Pe = 5000). The flow pathway becomes more tortuous due to the growth of CO_2 bubbles. The distribution of the shear rate ($\dot{\gamma}$) near the rock surface is calculated from the velocity fields in A–D, as shown in figures labeled with 1, 2, 3, 4, respectively. (e) Probability density functions for the velocity of the liquid, $p(V)$ at t = 1, 20, 40, and 60 s, derived from PIV data. (f) Evolution of the reactive surface area (A_{react}) and shear force ($f_s = \mu\dot{\gamma}A_{react}$) on the rock surface. Due to the shielding and weakening effect of CO_2 bubbles, A_{react} and f_s decrease from 20 s to 60 s (b–d).

$$R_{CD} \sim C_{avg} \cdot A_{react} \quad (2a)$$

$$R_{ME} \sim \tau_b^{1.5} \cdot A_{react} \quad (2b)$$

where A_{react} is the reactive surface area exposed to the acid, C_{avg} is the average acid concentration in the channel (driving chemical erosion), and τ_b is the shear stress on the rock

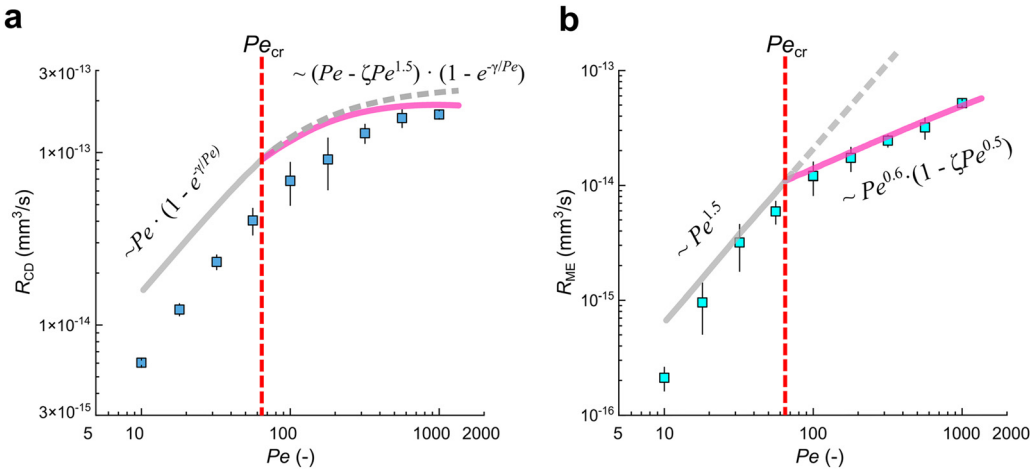


Fig. 9 Theoretical scaling relationship between erosion rates and Pe. (a) R_{CD} and (b) R_{ME} show distinct scaling behavior across regimes. A critical Pe separates transport-limited (gray solid line) and reaction-limited (purple line) regimes. Bubble-induced shielding and shear weakening reduce the slope of R_{ME} at high Pe. Dashed gray lines represent models neglecting bubble effects. Error bars reflect variability from repeated experiments (n = 3).

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surface (driving mechanical erosion). The exponent 1.5 in eqn (2b) accounts for the weakened cohesion of non-reactive minerals—such as dolomite and clay—due to calcite dissolution in our limestone samples.[58]

To derive the erosion scaling relationship, we estimate three parameters, A_{react}, C_{avg}, and τ_b, under both transport-limited and reaction-limited conditions. In the transport-limited regime, the entire rock surface is exposed to reactive fluid, and thus $A_{react} = 2Lh_0$, where L is the channel length and h_0 is the initial channel aperture. The average concentration C_{avg} is derived from the 1D advection-diffusion-reaction equation in a channel, $C_{avg} = C_{in}Pe(1 - \exp(-\gamma/Pe))/\gamma$, where C_{in} is the inlet acid concentration. Shear stress τ_b is linearly related to the flow velocity gradient in the liquid film and is thus represented as $\tau_b \sim Pe$. Substituting these expressions into eqn (2a) and (2b), we obtain the scaling laws for the transport-limited regime:

$$R_{CD} \sim Pe\left(1 - e^{-\frac{\gamma}{Pe}}\right) \quad (3a)$$

$$R_{ME} \sim Pe^{1.5} \quad (3b)$$

In contrast, the reaction-limited regime is strongly influenced by CO_2 bubble formation, which alters both the available reactive surface and the local flow structure. Due to bubble shielding, the effective reactive surface is reduced. We estimate this effect as: $A_{react} = 2(1 - \zeta Pe^{0.5})Lh_0$, where ζ is an empirical parameter capturing the shielding efficiency. The empirical parameter was obtained by establishing a quantitative relationship between the size of the attached CO_2 bubbles and the amount of acid consumed (Fig. S10). The detailed derivation of this parameterization is provided in Text S2. C_{avg} remains as in eqn (3a). However, shear stress τ_b is now affected by localized flow disturbances caused by bubble dynamics, including eddy formation, interface motion near the three-phase contact line, and constrictions in the flow path. Analytical derivation of τ_b under such conditions is challenging. Therefore, we perform pore-scale numerical simulations to capture the evolution of τ_b under varying bubble heights h_b and Pe (Text S3 and Fig. S11). Results show an empirical power-law relationship: $\tau_b \sim Pe^{\xi}$, with $\xi = 0.4$. Substituting the modified A_{react} and τ_b into eqn (2), we derive the scaling laws for the reaction-limited regime:

$$R_{CD} \sim \left(P_e - \zeta Pe^{1.5}\right)\cdot\left(1 - e^{-\frac{\gamma}{Pe}}\right) \quad (4a)$$

$$R_{ME} \sim Pe^{0.6}\cdot(1 - \zeta Pe^{0.5}) \quad (4b)$$

The detailed derivation of the scaling law is provided in Text S2. These scaling relationships (eqn (4a) and (4b)) show excellent agreement with experimental results (Fig. 9), particularly in capturing the shift in erosion behavior under multiphase flow. For comparison, we also plotted scaling laws assuming no multiphase effects (dashed lines in Fig. 9), where $\tau_b \sim Pe$ and $A_{react} = 2L_0h_0$ are applied uniformly across regimes.

Our findings indicate that chemical dissolution rate (R_{CD}) is only weakly impacted by multiphase effects in the reaction-limited regime. CO_2 bubbles reduce the exposed reactive surface area, thus lowering the dissolution rate. However, they also constrict the flow channel, enhancing mixing and reactant transport near the interface. These two competing mechanisms tend to offset each other (Fig. 8e), resulting in a negligible net change in R_{CD} (Fig. 9a). In contrast, R_{ME} is more sensitive to multiphase flow. CO_2 bubbles significantly reduce both the effective shear stress and the surface area available for detachment of non-cohesive particles. Besides, the erodibility of the undissolved mineral particles also depends on the chemical dissolution rate of the fast-reacting calcite.[37] As a result, R_{ME} is substantially lower in the reaction-limited regime compared to predictions that neglect multiphase effects (Fig. 9b). This highlights the dominant influence of bubble dynamics on both chemical dissolution and mechanical erosion. Overall, the derived scaling laws successfully capture the erosion dynamics under both transport- and reaction-limited conditions, particularly when considering the effects of multiphase flow. These results emphasize the need to account for gas–liquid phase interactions and local flow restructuring when predicting erosion behavior in reactive multiphase systems.

Although the theoretical framework and scaling analysis are based on experiments using a geometrically simple fracture and a relatively homogeneous limestone sample, the developed methodology—combining real-rock microfluidics, high-resolution imaging, and quantitative erosion rate measurements—can be extended to more complex systems. For instance, intersecting fractures, variable mineral compositions, and heterogeneous grain textures can be incorporated to explore erosion behavior under broader geochemical and flow conditions. These investigations are promising directions for future research, particularly for upscaling erosion rates in natural and engineered fractured media.

4. Conclusion and perspectives

This study reveals how multiphase flow conditions induce heterogeneous erosion patterns under varying hydrodynamic regimes. By integrating micro-PIV, fluorescence microscopy, and ion chromatography, we developed an experimental platform capable of *in situ* imaging of solid–liquid–gas interfaces and flow fields at high spatiotemporal resolution within real-rock microfluidic systems. This setup enables simultaneous observation of rock surface evolution, multiphase fluid interfaces, and mineral particle detachment, allowing for coupled analysis of chemical dissolution and mechanical erosion—processes traditionally studied in isolation. The accuracy and versatility of our method have been validated, demonstrating its capability to quantify both erosion regimes across a range of flow conditions. Furthermore, our rock-based microfluidic fabrication approach exhibits good reproducibility

across lithologies and hydrodynamic conditions, making the platform adaptable to rock samples containing heterogeneous minerals and complex geometries such as fracture networks and dual-porosity media. This versatility highlights its potential for studying broader phenomena such as coupled erosion–precipitation dynamics, opening new opportunities to investigate chemical, physical, and biological processes at the pore-scale.

Applying this platform to limestone erosion, we identify a regime transition driven by bubble mobility with increasing flow velocity. Through experiments and theoretical analysis, we determined a threshold flow rate separating two erosion regimes: transport-limited and reaction-limited. In the transport-limited regime ($Pe < Pe_{cr}$), CO_2 bubbles remain immobile, confining flow to thin liquid films. Here, the erosion rate is strongly governed by flow dynamics, and increasing velocity significantly enhances erosion. In the reaction-limited regime ($Pe > Pe_{cr}$), mobile CO_2 bubbles form and attach to the rock surface, shielding reactive areas and reducing local shear stress. This bubble-induced shielding mitigates the velocity-driven increase in the erosion rate, resulting in a less pronounced growth with the flow rate. We have derived scaling relationships for chemical dissolution (R_{CD}) and mechanical erosion (R_{ME}) rates as functions of Pe in both regimes. In the transport-limited regime, scaling slopes remain constant, reflecting direct hydrodynamics control on erosion rates. In contrast, the reaction-limited regime exhibits decreased slopes, attributed to the dual effects of reactive surface area shielding and shear stress reduction on rock erosion. Notably, R_{CD} is less affected due to enhanced solute transport through the narrowed flow paths created by bubbles. However, R_{ME} shows a marked slope reduction, reflecting the more direct impact of bubble shielding and weakening on mechanical particle detachment. These insights help resolve longstanding inconsistencies in multiphase erosion studies, clarifying that the observed effects of multiphase flow are strongly dependent on whether the flow exceeds a critical Pe threshold.

Accurate erosion rate estimation is essential for reliable continuum-scale reactive transport modeling. Previous studies have reported large discrepancies—up to several orders of magnitude—between laboratory-measured dissolution rates and model predictions, largely due to differences in flow conditions, mineralogical heterogeneity, and reactive surface areas.[68–71] Critically, mechanical erosion is often omitted in these models, despite its significant role in reshaping flow pathways in natural rocks.[16,19,72] Its exclusion may lead to significant underestimation of the overall erosion and fracture geometry, depending on the mineral content and distribution. Our work addresses this gap by quantifying the concurrent evolution of both chemical and mechanical erosion within a realistic geological context. Moreover, the proposed imaging and analysis framework offers a versatile, scalable toolset for exploring erosion dynamics in structurally and mineralogically heterogeneous systems with varying fluid compositions.

Author contributions

Conceptualization: C. X. Z., R. H., B. L.; funding acquisition: C. X. Z., R. H., B. L., Y. F. C.; methodology: C. X. Z., R. H., B. L., H. D.; resources: R. H., B. L., Y. F. C.; software: C. X. Z., H. D., Z. Y.; investigation: C. X. Z., R. H., B. L.; visualization: C. X. Z., R. H., B. L., Z. Y.; supervision: R. H., Y. F. C.; writing – original draft: C. X. Z.; writing – review & editing: R. H., B. L.

Conflicts of interest

The authors declare no conflict of interest.

Data availability

Data for this article, including the relevant quantitative original data, are available at figshare at https://doi.org/10.6084/m9.figshare.27931161.v2.

The measurement procedures for erosion rate and the derivation process of the erosion rate scaling are detailed in the supplementary information. See DOI: https://doi.org/10.1039/d5lc00773a.

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